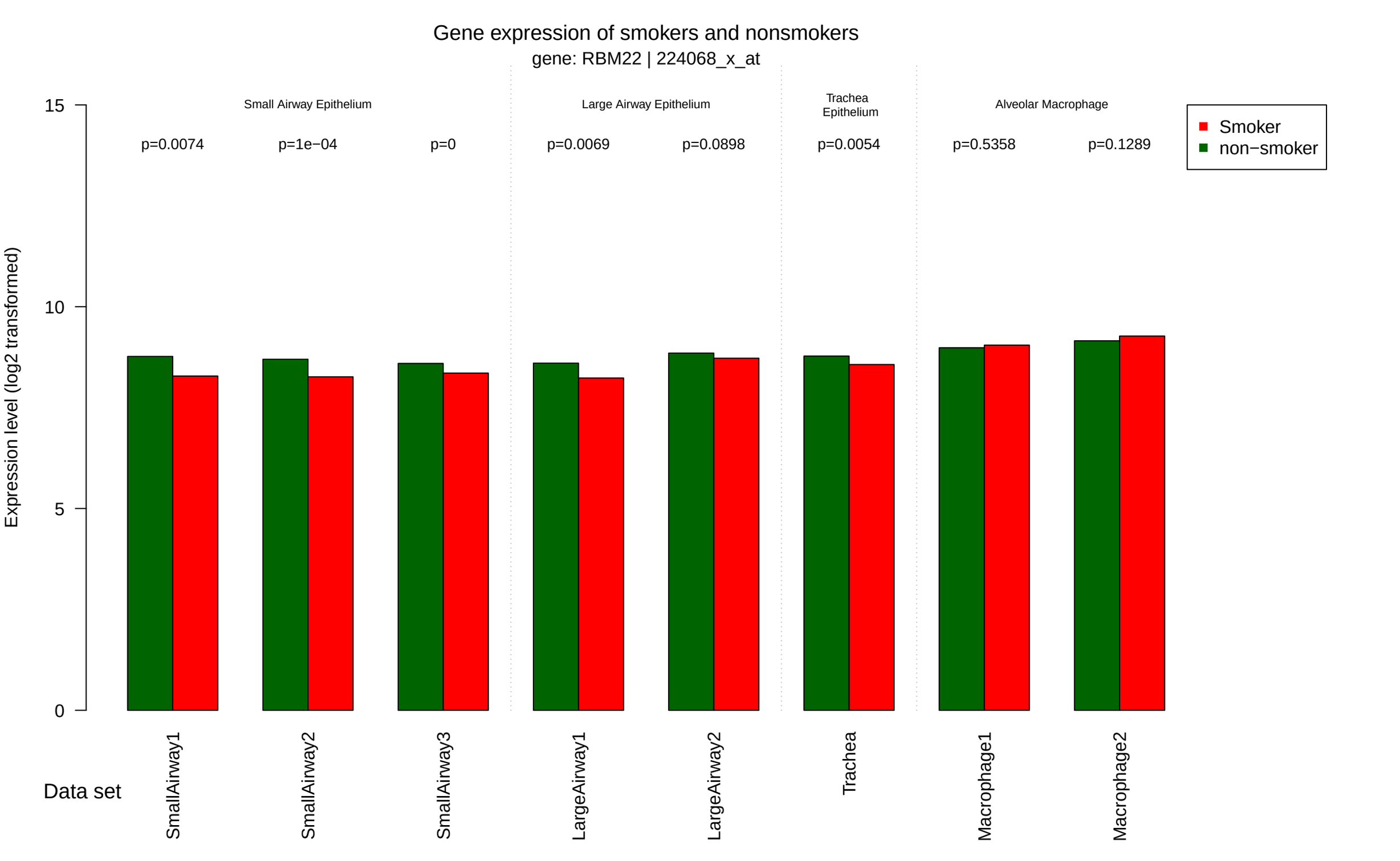
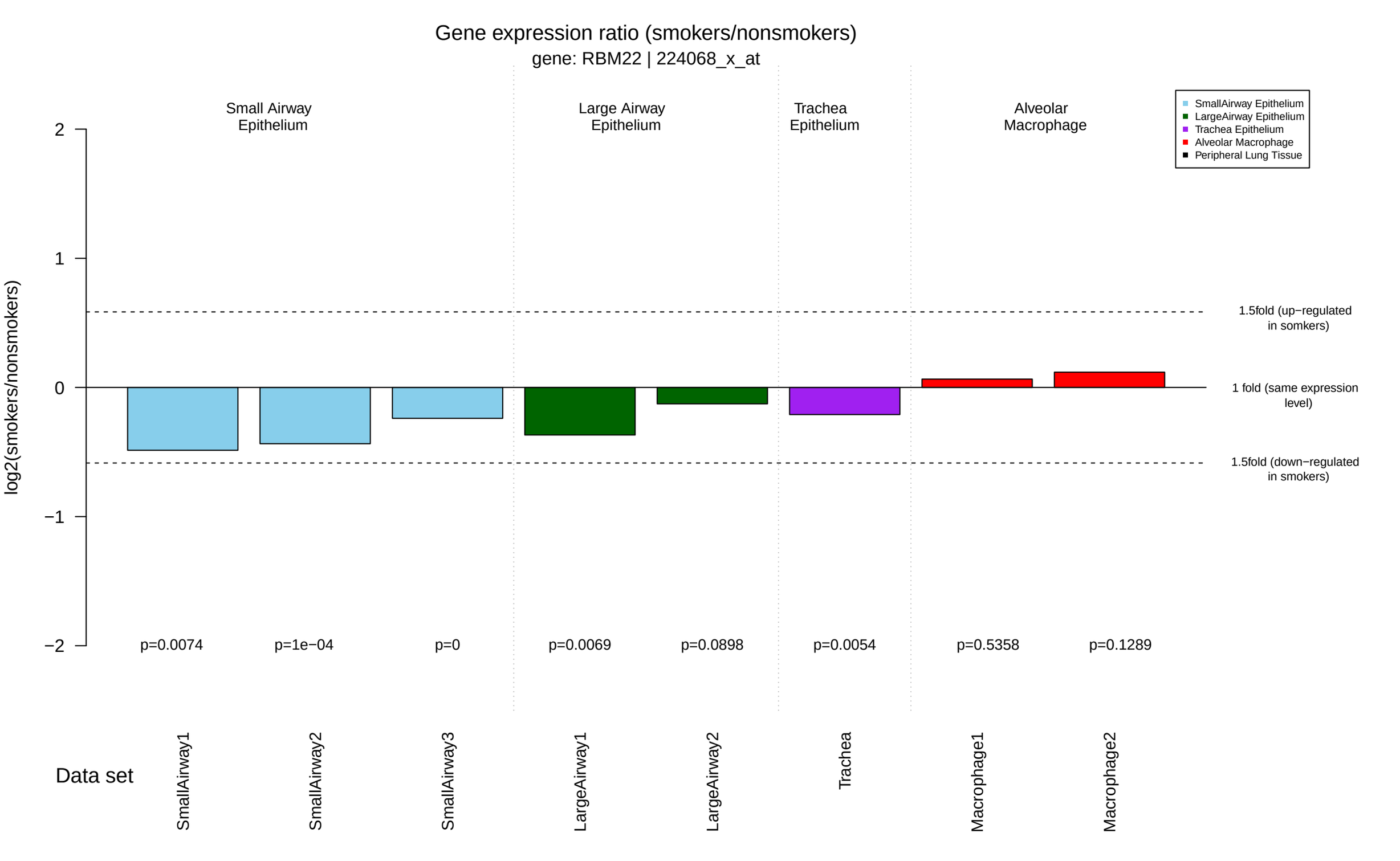
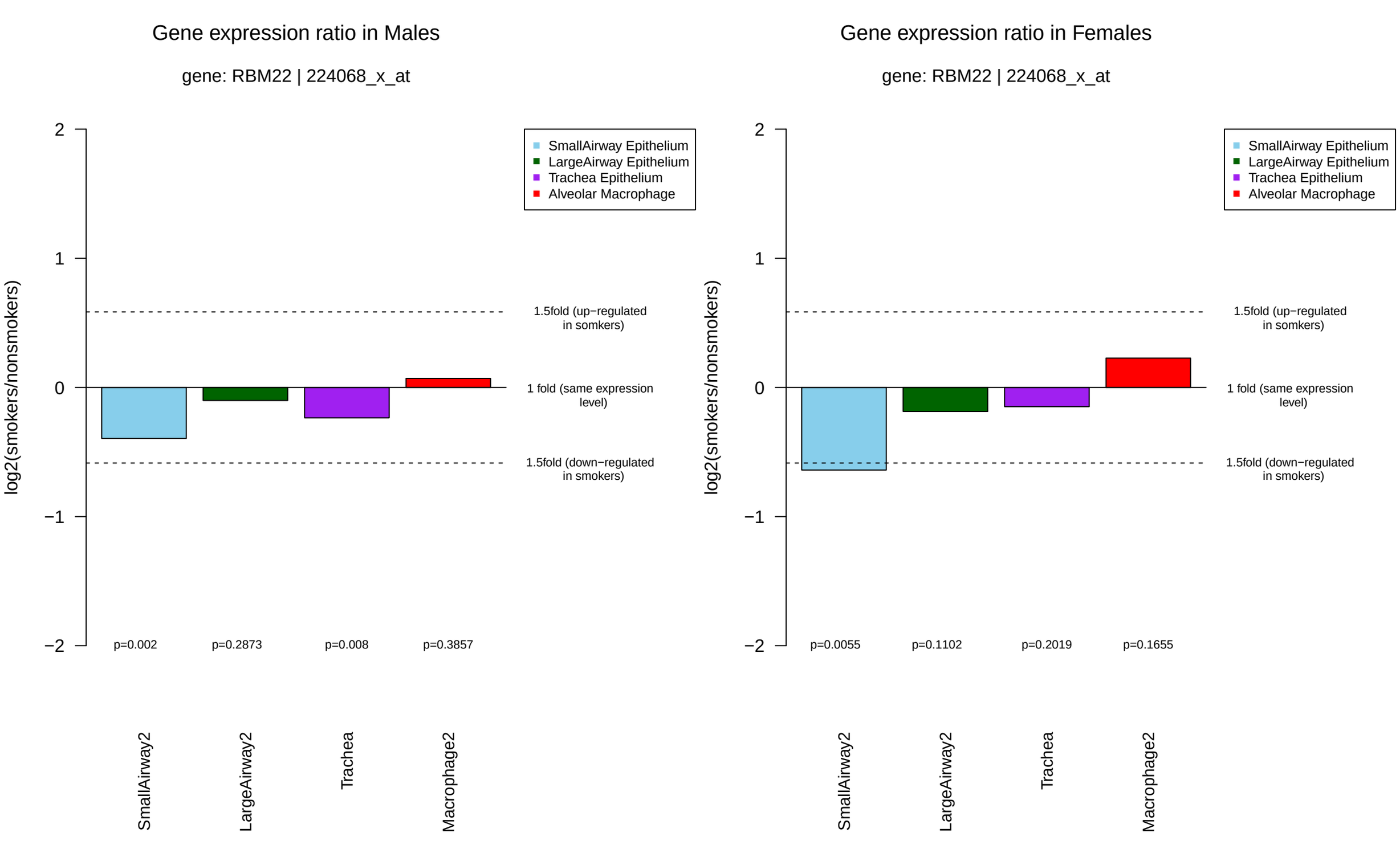
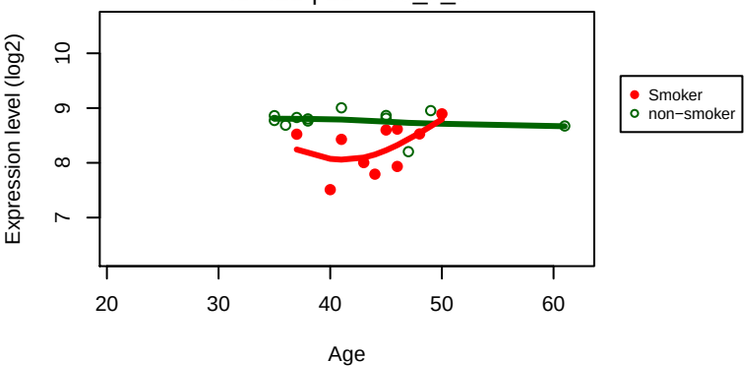
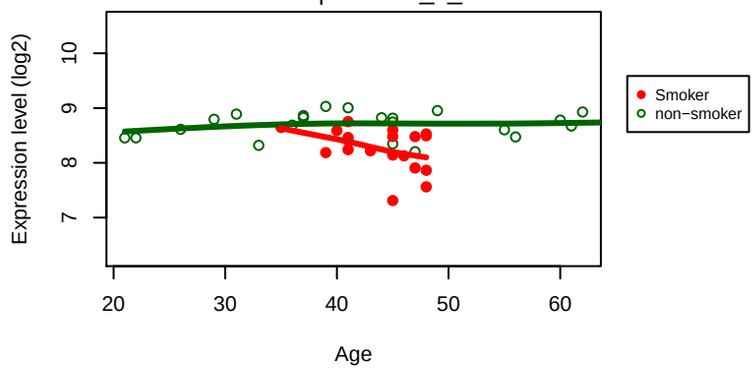
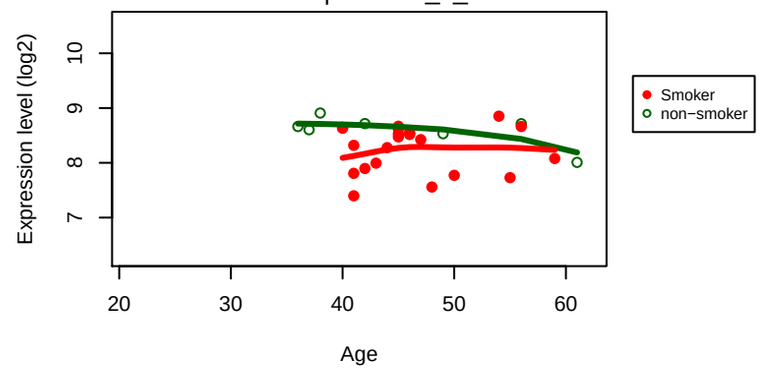

Gene expression of smokers and nonsmokers
gene: RBM22 | 224068_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0074
p=1e−04
p=0
p=0.0069
p=0.0898
p=0.0054
p=0.5358
p=0.1289
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

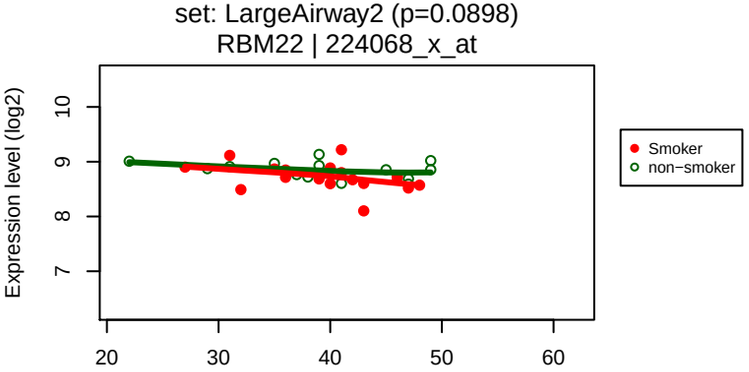

Gene expression ratio (smokers/nonsmokers)
gene: RBM22 | 224068_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0074
p=1e−04
p=0
p=0.0069
p=0.0898
p=0.0054
p=0.5358
p=0.1289
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

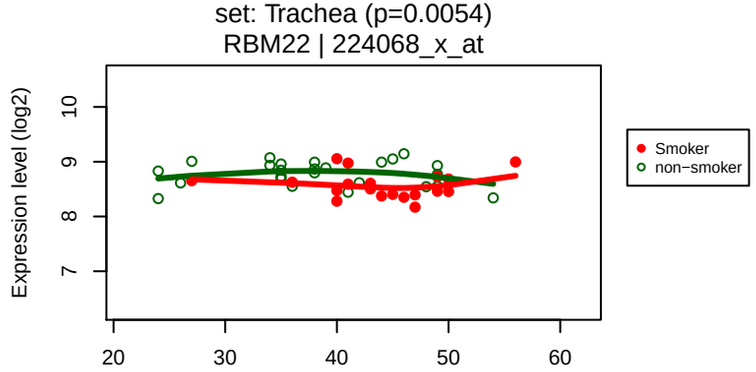

Gene expression ratio in Males
gene: RBM22 | 224068_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.002
p=0.2873
p=0.008
p=0.3857
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RBM22 | 224068_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0055
p=0.1102
p=0.2019
p=0.1655
SmallAirway2
LargeAirway2
Trachea
Macrophage2

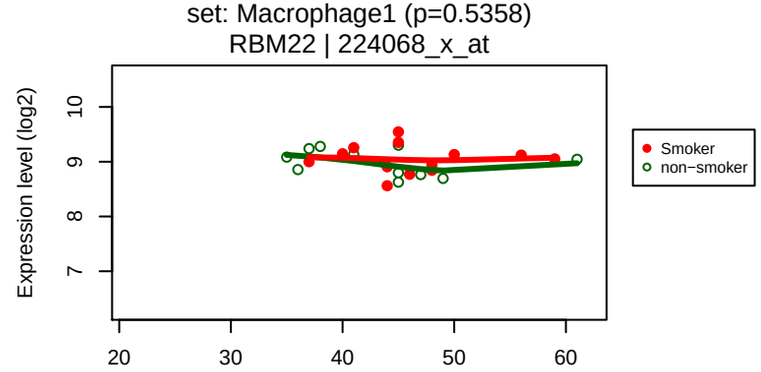

set: SmallAirway1 (p=0.0074)
RBM22 | 224068_x_at
Expression level (log2)
Age
Smoker
non−smoker

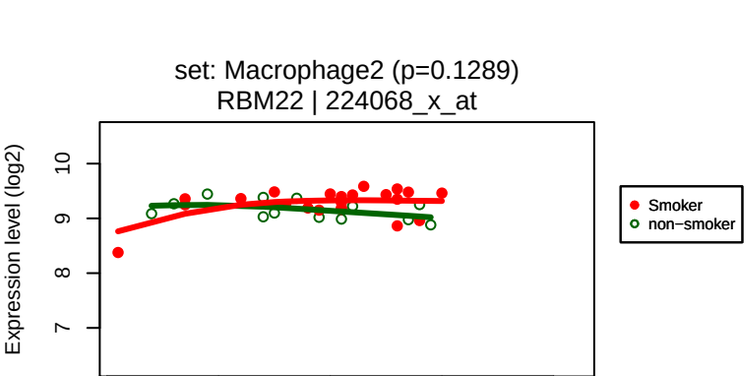

set: SmallAirway2 (p=1e−04)
RBM22 | 224068_x_at
Expression level (log2)
Age
Smoker
non−smoker

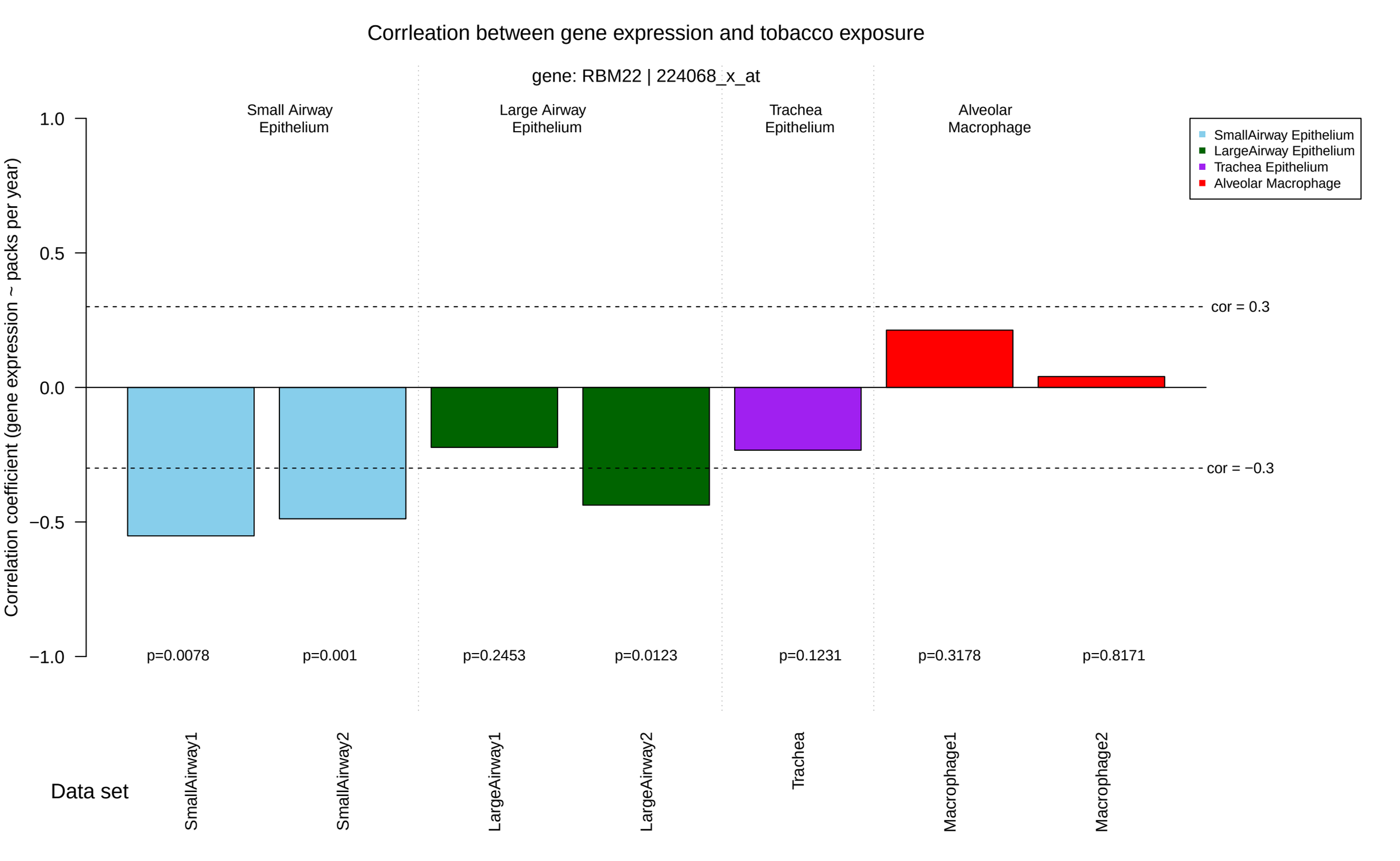

set: LargeAirway1 (p=0.0069)
RBM22 | 224068_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0898)
RBM22 | 224068_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0054)
RBM22 | 224068_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5358)
RBM22 | 224068_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1289)
RBM22 | 224068_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RBM22 | 224068_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0078
p=0.001
p=0.2453
p=0.0123
p=0.1231
p=0.3178
p=0.8171
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2